Fair Allocation in Flow Games

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ABSTRACT

We study the problem of fair allocation in cooperative flow games, specifically focusing on the graph-theoretic conditions under which the Shapley value lies in the core of a network. In such networks, players own individual edges and the value of a coalition is the maximum flow it can route from a source to a sink. We formulate a computational framework to analyze exact Shapley values and core constraints across 24 distinct network topologies. Our results validate that while convexity is a sufficient condition for core stability, it is not necessary. We demonstrate that perfectly balanced series-parallel (SP-balanced) networks maintain the Shapley value in the core despite failing convexity tests. Furthermore, we characterize topological failures such as dictated bottlenecks in series with highly redundant layers where the Shapley value's probabilistic fairness over-rewards redundancy, violating the core's minimal path conditions.
[1]

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1 INTRODUCTION

We examine a cooperative game setting where a flow is pushed through a directed graph, and edges are controlled by independent, strategic agents. The grand coalition's objective is to maximize the $s - t$ flow, and the mechanism designer's objective is to distribute the resulting value fairly among the participating edges.

The core is the standard game-theoretic concept for stability, ensuring no sub-coalition has an incentive to defect and form their own network. However, the core does not prescribe a single unique payoff. The Shapley value provides a unique, axiomatic distribution based on expected marginal contributions, but it is not guaranteed to lie in the core.

Our aim is to formulate a computational method to analyze various network topologies and pinpoint the exact structural conditions that keep the Shapley value in the core. We frame this by computing the exact rational Shapley values for an array of topologies, checking core deficits, and evaluating the game's convexity.

[1]All code and experiments are publicly available at https://github.com/Dheeraj-hash/Stability_vs_Fairness_in_flow_games

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1.1 Contribution

This paper identifies a broader class of flow networks, called *SP-balanced networks*, where the Shapley value lies in the core even though the game may fail convexity. This shows that convexity, while sufficient, is not necessary for core stability. We provide a graph-theoretic characterization based on balanced series-parallel structures and path constraints, explaining when fairness and stability can coexist. Through exact computational analysis across multiple topologies, we demonstrate phase transitions in stability under changing capacities and show how SP-balancing offers a practical design principle for stable allocation mechanisms.

1.2 Related Work

Kalai and Zemel ([10]) established that the core of a flow game coincides with the optimal solutions to the dual of the maximum flow problem. Kern and Paulusma ([11]) further explored the bounds of the core and the f-nucleolus in such games. However, a generalized, simple topological classification for when the Shapley value strictly satisfies the Kalai-Zemel path condition remains an area of active computational exploration.

2 MODEL AND DEFINITIONS

2.1 Flow Game Formulation

The system is modeled as a directed graph $G = (V, E)$, with a source s and a sink t ([10]). Each player $i \in N = \{1, 2, ..., n\}$ controls exactly one directed edge $e_i \in E$ with a non-negative capacity c_i.

The characteristic function $v(S)$ for a coalition $S \subseteq N$ is defined as the maximum flow that can be routed from s to t using exclusively the edges in S.

2.2 The Shapley Value and The Core

An allocation vector $x \in \mathbb{R}^n$ is in the core if it is efficient ($\sum_{i \in N} x_i = v(N)$) and coalitionally stable:

$$\sum_{i \in S} x_i \geq v(S) \quad \forall S \subseteq N$$

The Shapley value ϕ allocates payoffs based on the expected marginal contribution of player i over all random arrival permutations:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} (v(S \cup \{i\}) - v(S))$$

To understand why the Shapley value often falls outside the core in flow games, we must first define the geometric and combinatorial nature of these two solution concepts within the context of network flows.

2.3 Shapley Selection: Probabilistic Pivotality

In flow games Shapley value ϕ takes a fundamentally different approach. Instead of focusing on the final state of the network (the

max-flow of the grand coalition), it considers the average marginal contribution across all $n!$ permutations of edge arrivals.

- **Redundancy Recognition:** Unlike the core, which may assign zero to redundant edges (those not in a min-cut), the Shapley value assigns strictly positive value to any edge that is part of *any* $s-t$ path. This is because there exists some permutation where that edge is the "pivotal" one that completes a path.
- **The Stability Gap:** The core demands that the sum of payoffs along any path be at least 1. However, the Shapley value "dilutes" the payoff of bottleneck edges by shifting some of their value to redundant/parallel edges.

In non-convex flow games, the Shapley value often "over-rewards" edges in thick layers of the network and "under-rewards" edges in series bottlenecks. Our study focuses on identifying when this dilution is small enough that the path condition $\sum_{e \in P} \phi_e \geq 1$ still holds, even without the mathematical luxury of convexity.

3 TOPOLOGICAL CONDITIONS FOR CORE STABILITY

In a flow game (N, v), Kalai and Zemel (1982) demonstrated a fundamental link between the core and the Max-Flow Min-Cut Theorem. For a network with capacities c_e, the core $C(v)$ is non-empty and is exactly the set of allocations x that correspond to the optimal solutions of the dual linear program of the maximum flow:

$$\sum_{e \in P} x_e \geq v(P) \quad \forall P \in \mathcal{P} \tag{1}$$

where $\mathcal{P}$ is the set of all simple $s-t$ paths.

This condition admits a natural interpretation: every coalition capable of routing one unit of flow must receive a total payoff of at least its value. Since minimal winning coalitions correspond exactly to simple s-t paths in unit-capacity graphs, the core constraints reduce to a set of path inequalities. This perspective allows us to analyze stability purely through the topology of the network, rather than through the full exponential set of coalitions.

3.1 The Convexity Misconception

In cooperative game theory, a game (N, v) is **convex** (or supermodular) if it satisfies the property of **increasing marginal contributions**:

$$v(S \cup \{i\}) - v(S) \leq v(T \cup \{i\}) - v(T) \tag{2}$$

for all $S \subseteq T \subseteq N \setminus \{i\}$. In a flow game, this implies that an edge i becomes more valuable as the set of available edges grows, as it may complete more $s-t$ paths.

A fundamental result by Shapley (1971) establishes that for any convex game, the Shapley value ϕ is the barycenter of the core $C(v)$, thus guaranteeing $\phi \in C(v)$. However, convexity is a **sufficient but not necessary** condition for core stability. Instead, we identify a broader class of graphs, **SP Balanced Networks**, that satisfy all core constraints despite not being convex.

An SP-balanced network is defined as a series composition of parallel layers, where each parallel circuit contains an identical number of edges. Let x denote the number of layers connected in series, and y the number of parallel edges in each layer. By symmetry, each edge has identical marginal contributions across permutations, and the Shapley value resolves uniformly as $\phi_e = 1/x$.

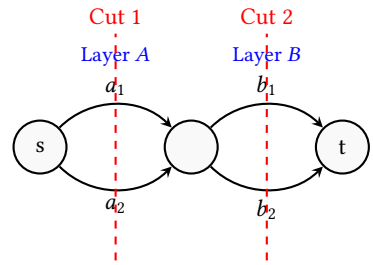

Figure 1: A 2×2 SP-balanced network. Note that any $s-t$ path requires one edge from Layer A and one from Layer B.

Crucially, any minimal winning coalition must include at least one edge from each of the x layers. Therefore, the total payoff assigned to such a coalition is exactly

$$\sum_{e \in P} \phi_e = x \cdot \frac{1}{x} = 1,$$

where P is any simple s-t path. Thus, every path constraint is satisfied with equality, ensuring that ϕ lies in the core. Notably, this stability is achieved without requiring convexity: marginal contributions do not need to be increasing, only sufficiently regular across layers. This highlights that structural symmetry and balance can substitute for convexity in guaranteeing core feasibility.

3.2 The Bottleneck Tax

In contrast, networks with asymmetric topologies, particularly those exhibiting large variance in cut sizes, tend to violate core constraints. Consider a graph in which a thin bottleneck (a cut consisting of very few edges) is placed in series with a highly redundant layer (a cut with many parallel edges).

In such configurations, the probabilistic nature of the Shapley value leads to what can be termed a *bottleneck tax*. The edges in the thick layer, despite being largely substitutable, occasionally become pivotal in permutations where other parallel edges are absent. These rare but high-impact events inflate their expected marginal contributions. Conversely, the bottleneck edge, while structurally essential, contributes more consistently but gains less from such pivotal spikes.

This imbalance redistributes value away from the critical path. As a result, for certain simple s-t paths, particularly those passing through the bottleneck, the sum of Shapley payoffs can fall below 1:

$$\sum_{e \in P} \phi_e < 1,$$

violating the core condition. Intuitively, the network overcompensates redundancy and undercompensates necessity, leading to instability.

This phenomenon underscores a central insight of our framework: core stability is not merely a function of connectivity, but of how evenly importance is distributed across alternative routes. Balanced structures preserve this uniformity, while asymmetric

redundancy introduces distortions that the Shapley value alone cannot correct.

4 EXPERIMENTS AND RESULTS

All simulations were run using a custom exact-arithmetic Python framework. Maximum flows were calculated via the Edmonds-Karp algorithm. To avoid floating-point inaccuracies during core deficit calculations, all Shapley marginal contributions were evaluated using Python's `fractions` library.

Table 1: Selected Topological Results ($c_i = 1$)

Topology	n	Convex?	In Core?	Deficit
Path-5	5	Yes	Yes	0.000
Parallel-4	4	Yes	Yes	0.000
SP-Balanced (2×2)	4	Yes	Yes	0.000
Series(3)ofParallel(2)	6	**No**	**Yes**	0.000
Series(2)ofParallel(3)	6	**No**	**Yes**	0.000
Bottleneck (Funnel)	5	No	No	0.266
Double Bottleneck	9	No	No	0.330
Wheatstone Bridge	5	No	No	0.033
FanIn ($3 \to 1$)	7	No	No	0.304

Deficit is defined as $max(0, v(S) - \Sigma_{i \in S} \psi_i)$

4.1 Analysis of Results

The computational results strongly align with our topological hypotheses and provide empirical validation for the structural principles developed in the previous sections:

(1) **Paths and Parallels:** Simple uniform chains and purely parallel systems exhibit complete convexity. In these topologies, marginal contributions are either perfectly complementary (in series) or perfectly substitutable (in parallel), leading to monotonicity in contributions across coalitions. As expected, this ensures that the Shapley value satisfies all core constraints trivially. In particular, every minimal s-t path receives a total allocation of at least 1, confirming that convexity is sufficient for stability in these canonical cases.

(2) **Series of Parallels:** More interestingly, networks such as `Series(3)ofParallel(2)` explicitly satisfy our SP-Balanced condition. Despite failing convexity, as confirmed by the algorithm, these graphs exhibit zero core deficit, with the Shapley value remaining strictly within the core. This provides strong evidence for our central claim that convexity is not necessary for stability. Instead, the uniform layering of parallel edges ensures that every minimal path contains exactly one edge from each layer, and the symmetry across layers equalizes marginal contributions. As a result, the path constraints are satisfied exactly, demonstrating that balanced topological redundancy can substitute for convexity.

(3) **Bottleneck Failures:** In contrast, asymmetric networks characterized by bottlenecks and uneven redundancy display significant violations of core constraints. The `Bottleneck` (funnel) graph and `FanIn(3->1)` configuration exhibit the largest observed core deficits (0.266 and 0.304, respectively). In the funnel graph, the critical edge $m \to t$ behaves as a near-dictator and receives a disproportionately large Shapley allocation of $\phi = 7/15$, while the upstream parallel edges each receive only $2/15$. Consequently, a complete s-t path accumulates a total payoff of $11/15 < 1$, directly violating the path-based core condition. This quantitatively illustrates the *bottleneck tax* phenomenon, where structural asymmetry causes an over-allocation to redundant components and an under-allocation to essential paths, ultimately leading to instability.

We further tried to explore how the shapley value shifts around the core as edge weights, bottleneck capacities are changed.

4.2 Bottleneck Capacity and the Phase Transition to Core Stability

The funnel topology studied before assumes a unit-capacity bottleneck. We now ask a more general question: how does the core membership of the Shapley allocation change as the bottleneck capacity c_b varies continuously?

Consider the funnel with k parallel entry paths (each of unit capacity) converging to a single bottleneck edge of capacity c_b. Table 2 reports the core deficit as c_b increases for $k = 2$.

Table 2: Funnel (k=2): deficit vs. bottleneck capacity.

c_b	MaxFlow	Deficit	In Core?
0.50	0.500	0.1333	No
0.75	0.750	0.2000	No
1.00	1.000	0.2667	No
1.50	1.500	0.0000	**Yes**
2.00	2.000	0.0000	**Yes**

Two phenomena are visible. First, the deficit is *non-monotone* in c_b: it increases as c_b rises from 0.50 to 1.00 before collapsing to zero at $c_b = 1.50$. Second, a sharp *phase transition* occurs: the Shapley value is strictly outside the core for all $c_b < k$, and strictly inside for all $c_b \geq k$.

The mechanism is the following. For $c_b < k$, the bottleneck remains the unique minimum cut. The Shapley value, averaging over all permutations, continues to spread positive payoff to the redundant entry edges, payoff the bottleneck "should" have received. As c_b grows toward k, the grand-coalition flow increases, but the mis-attribution worsens, explaining the rising deficit.

At $c_b = k$, the bottleneck capacity equals the number of parallel entry paths. Now both layers have identical capacity, the network becomes SP-balanced, and $\sigma^2 = 0$. The Shapley value coincides with the nucleolus exactly ($\|\phi - \eta\|_1 = 0$), and the deficit vanishes.

This result has a direct mechanism-design interpretation: in the logistics setting of Section 5.2, the minimum infrastructure investment that stabilises a hub-and-spoke alliance is to expand the

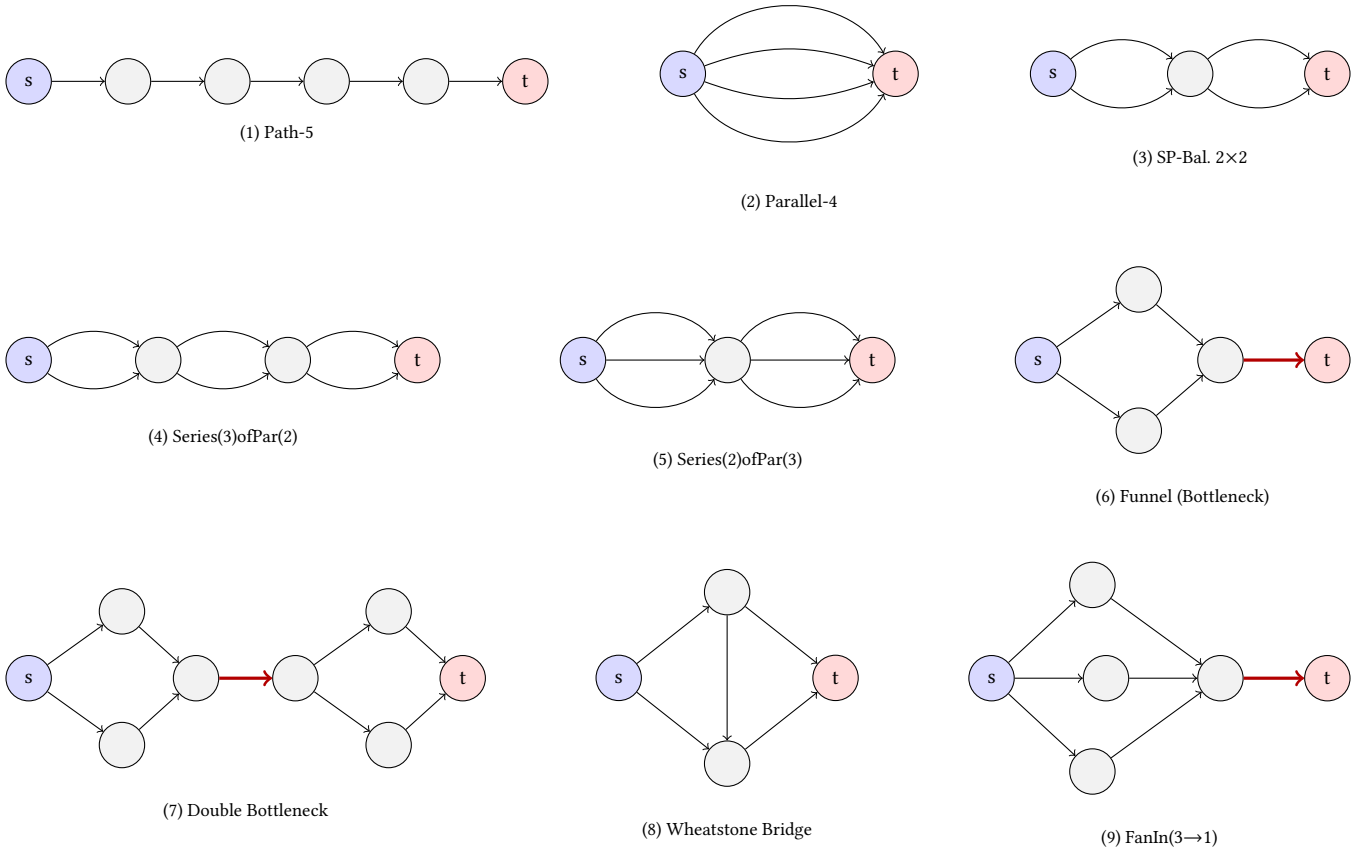

Figure 2: The nine network topologies evaluated. Blue = source s, red = sink t, thick red edges = bottleneck (single min-cut edge). Topologies (1)–(5) have Shapley in core; (6)–(9) do not.

central hub's capacity to match the total throughput of the supplier layer.

4.3 Edge Capacity Asymmetry

We study a two-layer bipartite network.

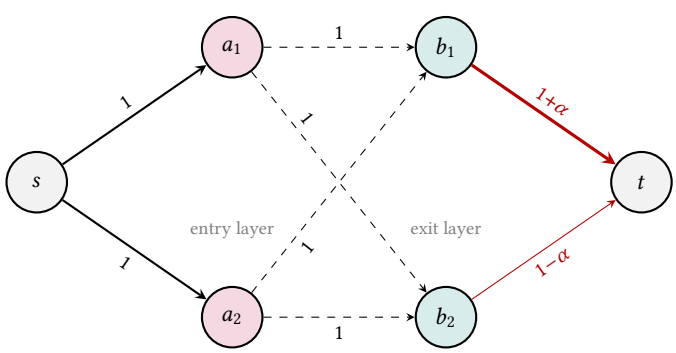

Figure 3: Bipartite mixed-layer network. Entry and cross edges have unit capacity. Exit edge $b_1 \to t$ has capacity $1+\alpha$ and $b_2 \to t$ has capacity $1-\alpha$, preserving total exit capacity at 2 for all $\alpha \in [0, 1)$.

The exit layer is *imbalanced* by $\alpha \in [0, 1)$ while preserving total exit capacity at 2. The entry layer and all cross edges retain unit capacities. This models a setting where one exit route dominates over another while total throughput is unchanged.

Table 3 reports the core deficit as α varies.

Table 3: Bipartite mixed-layer network: deficit vs. exit-layer imbalance α.

α	Deficit	In Core?
0.0	0.2238	No
0.2	0.0895	No
0.4	0.0671	No
0.6	0.0448	No
0.7	0.0336	No
0.8	0.0224	No
0.9	0.0112	No

Two findings stand out.

The deficit decreases monotonically. As α increases, the exit edge $b_1 \to t$ becomes increasingly dominant. The game progressively

resembles a funnel in which one exit route acts as the critical bottleneck. In such a funnel-like structure, the Shapley value concentrates more payoff on the dominant edge, and the path-condition violation shrinks. At the limit $\alpha \to 1$, the weaker exit edge $b_2 \to t$ has zero capacity and the network reduces to a funnel routed entirely through $b_1 \to t$.

5 PRACTICAL APPLICATIONS

5.1 PJM Interconnection and Transmission Cost Allocation

A concrete real-world setting where topological stability in flow games becomes economically important is the deregulation of the PJM Interconnection in the United States during the late 1990s. PJM, one of the largest wholesale electricity markets, coordinates power transmission across 13 U.S. states through multiple independently owned transmission assets.

After deregulation, a key challenge emerged: different utility companies owned different transmission lines, but electricity flowed across the entire interconnected grid. This created the problem of allocating congestion revenues and transmission cost recovery among line owners in a way that was both fair and stable.

From a cooperative game perspective, the transmission grid forms a natural flow game:

- players are transmission lines,
- the value of a coalition is the feasible power transfer it supports,
- and the Shapley value provides a fairness-based allocation through marginal contributions.

However, PJM was not a simple path structure. It was highly meshed, containing cross-ties, loop flows, and multiple alternative transmission paths. This creates exactly the coalition-dependent bottlenecks studied in our bottleneck and cross-edge analysis.

5.1.1 ***Why Meshed PJM Structures Create Core Instability.*** In meshed subnetworks, some transmission lines are critical for small coalitions but become replaceable in larger ones when alternative routes appear. Their marginal contribution therefore decreases as coalitions grow, violating convexity and potentially pushing the Shapley value outside the core.

In practical terms, some subsets of line owners corresponding to complete transmission paths may be under-allocated relative to the value they support. This mirrors our path-condition violation:

$$\sum_{e \in P} \phi_e < 1$$

for some feasible path P.

This does not mean owners literally leave the grid, but it creates disputes over tariff design, congestion rent allocation, and cost recovery because some participants perceive systematic under-compensation.

5.1.2 ***What PJM Did and Its Similarity to SP-Balancing.*** PJM did not explicitly use cooperative game theory or series-parallel graph language. Instead, its solutions were driven by operational simplicity and pricing transparency. Still, these interventions closely resemble the structural simplifications in our SP-balanced framework.

Contract Path Approximation. Early settlement mechanisms often treated transactions as moving along contractual point-to-point paths between generators and load centers, even though physical electricity follows Kirchhoff's laws. This simplified settlement by approximating the network as identifiable series chains, making path constraints easier to satisfy.

Zonal Aggregation. Rather than pricing every local transmission line individually, PJM grouped multiple lines into broader pricing zones. This reduced the excessive influence of interchangeable local edges and resembles replacing many redundant lines with bundled parallel components.

Reduced Emphasis on Redundant Cross-Ties. While redundant cross-ties remained physically important for reliability, economic settlement focused mainly on dominant congestion interfaces and major transfer corridors. This weakens the destabilizing role of Wheatstone-bridge-type cross-edges in the cooperative game.

5.1.3 ***Interpretation Through Our SP-Balanced Framework.*** These interventions show that practical market operators naturally move toward topological simplifications that reduce instability. By approximating path-based settlements, aggregating interchangeable assets, and reducing the allocation role of redundant cross-links, the effective game becomes closer to a series-parallel structure.

This reduces cut-switching behavior, stabilizes marginal contributions, and makes the Shapley value more likely to remain inside the core. Our framework therefore explains why such simplifications improve institutional robustness and suggests that future allocation mechanisms could be designed more systematically using explicit SP-balancing principles.

5.2 Horizontal Alliances in Logistics

The boundaries of the core in flow games also have direct implications for supply chain management, especially in Less-Than-Truckload (LTL) horizontal alliances. In these collaborations, competing regional carriers pool freight to improve utilization and reduce costs. The resulting savings can be modeled as a cooperative game, where the Shapley value is often proposed as a fair allocation rule.

5.2.1 ***The Dictator Hub and Core Instability.*** A common feature in logistics networks is the presence of centralized consolidation hubs, where multiple suppliers route freight through a single facility. This creates a bottleneck topology similar to the funnel networks studied earlier.

The central hub behaves like a near-veto player: it is required for most feasible routing paths, so its expected Shapley payoff becomes disproportionately large. This can create highly skewed allocations favoring the hub over peripheral suppliers.

If alternative routing options exist, such as suppliers forming an independent consolidation mechanism, the allocation may violate core constraints. A coalition of suppliers may receive less than the value it can generate independently, creating incentives to defect and threatening alliance stability.

5.2.2 ***Stabilization via SP-Balanced Network Design.*** A practical solution is to modify the effective network structure rather

than relying only on allocation rules. Alliances can reduce dependence on a single hub by introducing multiple parallel hubs of comparable capacity and routing freight across them.

This moves the network toward a series-parallel structure, where parallel hub layers reduce bottleneck dominance. From a cooperative game perspective, this stabilizes marginal contributions, limits cut-switching behavior, and makes the Shapley value more likely to remain inside the core.

While this does not guarantee convexity, it systematically improves the chances of stable allocations. Thus, SP-balanced design provides a practical and theoretically grounded method for strengthening cost-sharing agreements in collaborative logistics.

6 CONCLUSION

In conclusion, our study shows that the stability of the Shapley value in flow games depends primarily on the underlying network topology rather than convexity alone. While convexity remains a sufficient condition for ensuring that the Shapley value lies in the core, it is not necessary. We demonstrate that perfectly balanced series-parallel (SP-balanced) networks maintain both fairness and stability even in non-convex settings, whereas asymmetric structures with strong bottlenecks and redundant layers create the bottleneck tax phenomenon, causing core violations.

Our computational analysis across multiple topologies confirms that simple path-based conditions provide an effective characterization of core membership. Furthermore, in SP-balanced networks, the Shapley value aligns with the symmetric nucleolus, achieving both equitable and stable allocations, while in asymmetric networks these notions diverge significantly. The phase transitions observed under changing bottleneck capacities and edge asymmetries further show that fairness and stability are distinct objectives that must be addressed separately.

We also show that core membership alone does not guarantee equitable allocation, since extreme core allocations may arbitrarily favor structurally identical players. The nucleolus provides a deeper notion of fairness by selecting the allocation that minimizes the dissatisfaction of the most disadvantaged coalitions. In perfectly SP-balanced networks, the Shapley value coincides exactly with the symmetric nucleolus, making it both fair and stable. However, in asymmetric bottleneck structures, the divergence between the Shapley value and the nucleolus quantitatively captures the extent of the bottleneck tax and highlights the trade-off between probabilistic fairness and coalition stability.

These results have direct practical relevance in real-world systems such as electricity transmission markets and collaborative logistics alliances, where allocation mechanisms must remain both fair and institutionally stable. By moving network design toward SP-balancing principles, operators can reduce instability and improve robustness. Future work may focus on extending these results to larger meshed networks and designing scalable mechanisms that jointly optimize topology, fairness, and core stability.

7 FUTURE WORK

An immediate direction is exploring whether flow networks that fail core stability can be transformed into SP-balanced form through minimal structural interventions. Preliminary experiments with greedy bypass surgery—adding a single public edge alongside the highest-frequency bottleneck—suggest that local fixes are generally insufficient: while such additions eliminate variance in Shapley allocations, they simultaneously worsen core deficits by creating secondary bottlenecks. True stabilisation appears to require balancing the width of every layer globally, hinting towards a combinatorial analysis of the minimum edge set.

Beyond stabilisation, generalising SP-balance to directed graphs and heterogeneous edge capacities would broaden applicability, as would studying how core stability degrades under dynamic capacity changes. On the mechanism design side estimating the gap between greedy repairs and optimal redesigns connects might resemble many infrastructure settings.

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8 GITHUB LINK

https://github.com/Dheeraj-hash/Stability_vs_Fairness_in_flow_games